# ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

## Chapter 8: Introduction to Trigonometry (Trikonmiti)

### Key Concepts & Important Formulas:

* Trigonometric Ratios in Right Triangle:
- sin(theta) = Opposite / Hypotenuse
- cos(theta) = Adjacent / Hypotenuse
- tan(theta) = Opposite / Adjacent
- cosec(theta) = 1/sin, sec(theta) = 1/cos, cot(theta) = 1/tan

* Values of Trigonometric Ratios:
- sin(30) = 1/2, sin(45) = 1/sqrt(2), sin(60) = sqrt(3)/2, sin(90) = 1
- cos(30) = sqrt(3)/2, cos(45) = 1/sqrt(2), cos(60) = 1/2, cos(90) = 0
- tan(30) = 1/sqrt(3), tan(45) = 1, tan(60) = sqrt(3)

* Fundamental Trigonometric Identities:
1) sin^2(theta) + cos^2(theta) = 1
2) 1 + tan^2(theta) = sec^2(theta)
3) 1 + cot^2(theta) = cosec^2(theta)

### Solved Exam Examples & Practice:

Example: If sin A = 3/5, find cos A and tan A.
cos A = sqrt(1 - sin^2 A) = sqrt(1 - 9/25) = 4/5.
tan A = sin A / cos A = (3/5)/(4/5) = 3/4.